

Statistical Methods for Data Science (Fall 2025)

Mini Project 5

Instructions:

- Due date: Monday, Dec 10th, 2025.
- Total points = 20.
- Submit a typed report.
- Justify all steps and provide all relevant explanations.
- You can work on the project either individually or in a group of no more than two students. In the case of the latter, submit only one report for the group and include a description of the contribution of each member.
- It is OK to discuss the project with other students in the class (even those who are not in your group), but each group must write its own code and answers. If the submitted report (including code and answer) is similar (either partially or fully) to someone else's, this will be considered evidence of academic dishonesty, and you will be referred to appropriate university authorities.
- You must use the following template for your report:

Mini Project #

Name

Names of group members (if applicable)

Contribution of each group member

Section 1. Answers to the specific questions asked

Section 2: R code. Your code must be annotated. No points may be given if a brief look at the code does not tell us what it is doing.

1. Consider the data stored in `bodytemp-hearttrate.csv` on eLearning, containing measurements of body temperature and heart rate for 65 male (`gender = 1`) and 65 female (`gender = 2`) subjects.
 - (a) Do males and females differ in mean body temperature? Answer this question by performing an appropriate analysis of the data, including an exploratory analysis.
 - (b) Do males and females differ in mean heart rate? Answer this question by performing an appropriate analysis of the data, including an exploratory analysis.
 - (c) Is there a linear relationship between body temperature and heart rate? Please use a visualization tool to justify your hypothesis roughly.